

From: Doc Edge <edgem@usc.edu>
Sent: 2026-09-01 02:52:56 UTC
To: Alan Rogers <alan.rogers@hey.com>
Subject: new manuscript

Hi, Alan,

Great to hear from you! Thanks for sending this.

I agree with the main result. I haven't gone through and carefully translated between models, but both of my papers with Noah on this kind of thing have related statements about the difference in such a maximally-different trait in terms of within-population standard deviations growing in the square root of k . Confusingly, they are both 2015 papers because Human Biology was a year behind in their publications, but we have a 2015 haploid one based on the Edwards model in Studies in History and Philosophy of Science, and there it's immediately above equation 5. In the 2015 Human Biology one (which is for general ploidy and allows allele frequencies to vary among loci, though effect sizes are still the same magnitude across loci), the distributions in each population are in equation 54, and the root- k result follows if you take the difference and divide by the square root of the variance. I suspect that if we harmonize notation and translate assumptions between the papers (e.g. we don't have any environmental component in ours), the results will correspond in some sense—i.e. perhaps the ploidy = 2 special case of ours will equal the $h^2 = 1$ special case of yours?

In our paper, we aren't really advertising the results as talking about the maximum possible level of population differentiation on the trait; instead we are thinking about the level of population difference observed on a variable inspired by Edwards' classification procedure. I think you can also get the haploid result from Edwards' statements just above his "conclusion" section. Of course, Edwards isn't really thinking about a trait explicitly, and so he doesn't think of this in terms of an upper bound for a trait determined by a bunch of loci of equal effect, but I think it does show that the upper bound must be at least the one that follows from variable used for classification (under his assumptions).

As to the audience for publishing it, I'm not really sure. I don't really run into the claim that there **can't** be large trait differences between groups if average (or even if all) allele frequencies are small, but you run in overlapping but distinct circles from me, and presumably you see people saying such things. Certainly somebody who thought that such differences are impossible would be laboring under a misconception. I wonder whether a presentation that integrates this result with results related to your 1983 paper could reach more people and also be useful for trainees—it's interesting that the neutral distribution of trait differences (in standard deviation terms) doesn't change much in terms of its first couple of moments but nonetheless spreads out in terms of its bounds as k grows. Of course this is more redundant with your existing work and also can't be presented so directly as this.

It occurs to me that there may be some connections with results on local adaptation saying that most of the among-population variance in a polygenic trait under local adaptation resides in the covariance among loci rather in the locus-level differentiation. Berg and Coop 2014 point to a lot of this literature in their section on the Excess Variance Test, especially in the last subsection there <https://journals.plos.org/plosgenetics/article?id=10.1371/journal.pgen.1004412>. This, I think, relates to Edwards' statement (which he traces back to Fisher) about the power for classification coming from correlations among characters (loci).

-Doc

From: Alan Rogers <alan.rogers@hey.com> Sent: Monday, August 31, 2026 2:26 PM To: Doc Edge <edgem@usc.edu>
Subject: new manuscript

I've just written a 2-page note about population differences in allele frequencies vs quantitative characters. It's based on a model that is somewhat similar to the haploid version of the one you published in 2015, so I thought you might be

It asks this question: if F_{st} is small for a set of loci, how much does this constrain group differences in a quantitative character that these loci control? The answer is: very little if there are lots of loci.

This answer seems a bit obvious, and I was pretty sure it would turn out this way before I did the work. But it is my impression that it isn't obvious to everyone. So my first question to you is whether you agree. Is this of sufficient interest to be worth publishing.

Second, there is the issue of race. That is the primary context in which I see people claiming that there can be no important differences because F_{st} is small. But I am wary of that subject and have tried to steer clear. My second question is therefore: have I succeeded? I hope that this note will be useful to those who debate race without becoming central to that debate.

Also, I understand that everyone is busy. If you have the time to comment, I'll be grateful. If you don't, I completely understand.

Alan